

PYTHON FOR BEGINNERS · DAY 5

For Loops

Repeating with range()

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```
# count to 3
for i in range(1, 4):
    print(i)

# 1 2 3
```

TODAY'S GOALS

By the end of today, you can…

- Repeat code with **for** and **range()**
- Loop through each letter of a word
- Draw patterns with nested loops
- Use start, stop and step
- Add things up in a loop
- Choose between **for** and **while**

QUICK REVIEW · DAY 4

Same job, less code

while loop · 4 lines

```
count = 1
while count <= 5:
    print(count)
    count += 1
```

for loop · 2 lines

```
for count in range(1, 6):
    print(count)
```

No counter to create or update — and no infinite loops!

THE FOR LOOP

Once for each item in a sequence

```
for variable in sequence:
    statement
```

A **sequence** is a series of items — numbers, letters, and more.

Each round, the **loop variable** holds the current item.

Same rules as before: colon **:** and 4 spaces.

RANGE()

Making a sequence of numbers

Code	Numbers	Rule
range(5)	0, 1, 2, 3, 4	starts at 0
range(1, 6)	1, 2, 3, 4, 5	start, stop
range(0, 11, 2)	0, 2, 4, 6, 8, 10	third number = step
range(10, 0, -1)	10, 9, … 1	negative step counts down

The stop number is **never** included.

EXAMPLE

Counting down

Code

```
for i in range(5, 0, -1):
    print(i)

print("Blast off! 🚀")
```

Output

```
5
4
3
2
1
Blast off! 🚀
```

LOOPING THROUGH TEXT

A word is a sequence of letters

Count the vowels

```
word = "programming"
vowels = 0
for letter in word:
    if letter in "aeiou":
        vowels += 1
print(word, "has", vowels, "vowels")
```

Output

```
programming has 3 vowels
```

Each round, **letter** holds the next character: p, r, o, g…

THE ACCUMULATOR PATTERN

Adding things up in a loop

```
total = 0
for n in range(1, 101):
    total += n
print(total)  # 5050
```

1. Start at 0 **before** the loop
2. Add to it **inside** the loop
3. Use it **after** the loop

PATTERNS 🎨

Drawing with loops

Code

```
for row in range(1, 6):
    print("*" * row)
```

"*" * 3 gives *** — multiplying text repeats it!

A loop inside a loop is a **nested loop**: the inner loop runs fully for every outer round.

Output

```
*
**
***
****
*****
```

⚠️ MOST COMMON MISTAKE

The stop number is not included

```
range(1, 10)
# 1 … 9  ❌

range(1, 11)
# 1 … 10 ✅
```

To count to 10, the stop must be **11**.

WHICH LOOP?

for or while?

Use for when…

you know **how many times** to repeat

you go through **each item** of a sequence

print 10 times · each letter of a word

Use while when…

you repeat **until something happens**

you **don’t know** how many times

keep asking until the password is right

🧠 QUICK CHECK

Predict before you run!

Q1. Which numbers?

```
range(3, 8)
```

Q2. What is printed?

```
total = 0
for i in range(1, 5):
    total += i
print(total)
```

Q3. How many “Hi”?

```
for i in range(3):
  for j in range(2):
    print("Hi")
```

SUMMARY

What we learned today

Code	Meaning
for i in range(5):	repeat 5 times (0 to 4)
range(1, 11, 2)	start, stop, step → 1, 3, 5, 7, 9
range(10, 0, -1)	count down from 10 to 1
for letter in "hi":	loop through each letter
total += n	add up inside a loop
"*" * 5	repeat text → *****

💻 YOUR TURN

Practice time!

Open **exercises.ipynb** in the Day 5 folder and solve the 10 tasks, from ⭐ to ⭐ ⭐ ⭐.

⭐ Multiples of 5 up to 50

⭐ ⭐ Class average and highest score

⭐ ⭐ ⭐ Prime number finder 🔍

Next lesson: Strings 🔤